Mukuloth Badrinath

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.58
Alphores High School , Kothapally, Karimnagar DT TL, 57671	Class XII (Central Board of Secondary Education)	2023	82.40%
Alphores High School, BYPASS RD , Bommakal , Karimnagar	Class X (State Board of Secondary Education, Telangana state)	2021	10.00%

POSITIONS OF RESPONSIBILITY

Lead Production&Pronites : Milan General Championship | coordinator 2025 | Head 2026|

- Led the planning and execution of production, stage setup, dcor, and logistics for flagship events, ensuring seamless delivery and high participant engagement.
- Managed vendors, volunteers, and event execution to ensure seamless production and an engaging participant experience.
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Outreach Alumni Coordinator : Alumni cell IIT Hyderabad | Coordinator 2025-2026.

- Built and maintained relationships with alumni to strengthen engagement and institutional outreach.
- Coordinated alumni interactions, networking initiatives, and communication for institute events and programs.

Transport Coordinator : Inter IIT Sports Meet 2025

- Directed end-to-end transportation logistics for athletes, officials, and guests, ensuring safe, timely, and efficient mobility throughout the event.
- Coordinated with institute administration, transport vendors, and volunteer teams to optimize fleet operations and resolve real-time logistical challenges during a national-level sporting event.

Transport coordinator : Transport Department IIT Hyderabad | 2025-2026.

- Coordinated daily campus transportation operations, ensuring efficient scheduling, fleet management, and timely transport services for students, faculty, staff, and visitors.
- Collaborated with institute administration, transport vendors, and stakeholders to optimize transport logistics, address operational issues, and enhance service efficiency.

SKILLS

Languages : C, Python, R.

Technical skills : Bioinformatics & Molecular Visualization: VMD, PyMOL, IGV, BWA, SAM tools, GATK, FastQC, bcf tools.

Soft skills : Leadership, Team Management , Stakeholder Coordination, Time Management, Problem Solving, Public Speaking, Project Planning, Conflict Resolution.

PROJECTS

Next Generation Sequencing pipeline project :

- Developed and executed a complete **Next-Generation Sequencing (NGS)** variant-calling pipeline using **FastQC, BWA, SAMtools, GATK, and IGV** for quality control, sequence alignment, and somatic variant analysis.

- Performed genome alignment, BAM processing, variant calling with **Mutect2**, and variant visualization to identify and analyze genomic mutations using standard bioinformatics workflows.

Prime Number Distribution and the Riemann Hypothesis :

- Conducted computational analysis of prime number distribution, applying concepts from the Prime Number Theorem, the Riemann Hypothesis, and Möbius/Mertens functions to study mathematical patterns.

- Developed analytical visualizations and interpreted computational results to demonstrate interdisciplinary applications of number theory in cryptography, algorithms, and cybersecurity.

RELEVANT COURSES

Biotech : Basic Bioinformatics, Bio Nano Technology, Big Data Biology, Biostatistics.

Mathematics : Calculus, Probability & Random Variables, Linear algebra.

Computer science : C , Python, R , DSA.